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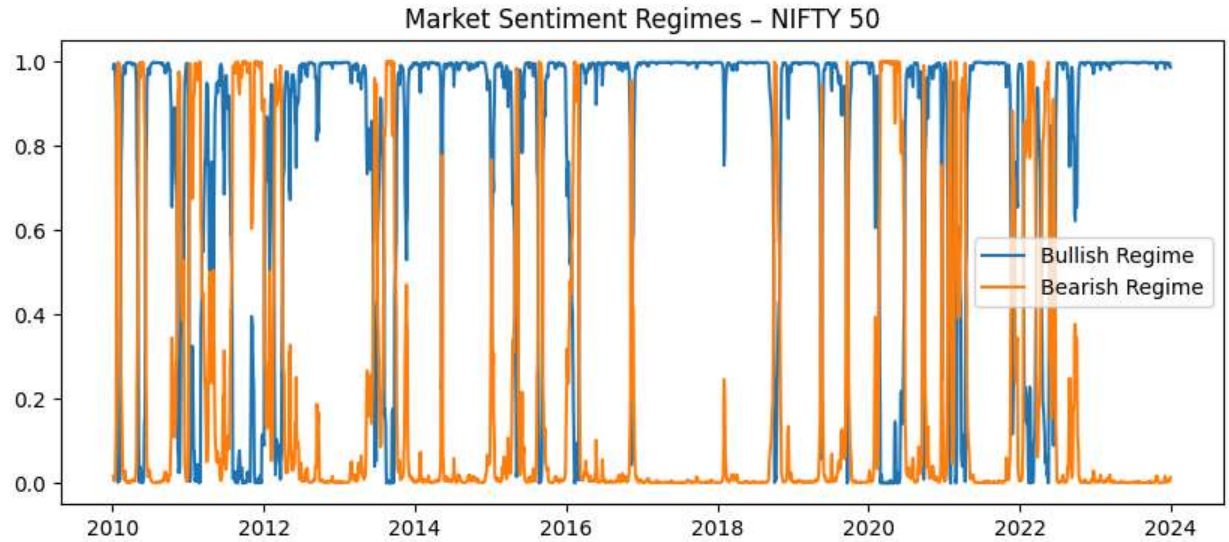
AIFBA EXPERIMENT : 04

```
# Exp no: 04 : analyzing market sentiment using the markov regime switching model
import yfinance as yf
import numpy as np
import pandas as pd
from statsmodels.tsa.regime_switching.markov_regression import MarkovRegression
import matplotlib.pyplot as plt
# Download NIFTY data
nifty = yf.download("^NSEI", start="2010-01-01", end="2024-01-01")
# Compute log returns
nifty['Returns'] = np.log(nifty['Close'] / nifty['Close'].shift(1))
returns = nifty['Returns'].dropna()
# Markov Regime Switching Model (2 regimes)
model = MarkovRegression(
 returns,
 k_regimes=2,
 trend='c',
 switching_variance=True
)
result = model.fit()
print(result.summary())
# Smoothed probabilities
probs = result.smoothed_marginal_probabilities
plt.figure(figsize=(10,4))
plt.plot(probs[0], label="Bullish Regime")
plt.plot(probs[1], label="Bearish Regime")
plt.legend()
plt.title("Market Sentiment Regimes – NIFTY 50")
plt.show()
```

```
/tmp/ipykernel_190/11649754.py:8: FutureWarning: YF.download() has changed argument
  nifty = yf.download("^NSEI", start="2010-01-01", end="2024-01-01")
[*********************100%***********************]  1 of 1 completed
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/base/tsa_model.py:473: Value
  self._init_dates(dates, freq)
                        Markov Switching Model Results
==============================================================================
Dep. Variable:                Returns   No. Observations:                 3433
Model:               MarkovRegression   Log Likelihood               11073.023
Date:                Thu, 12 Mar 2026   AIC                         -22134.046
Time:                        04:40:01   BIC                         -22097.199
Sample:                             0   HQIC                        -22120.883
                               - 3433
Covariance Type:               approx
                             Regime 0 parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.0008      0.000      5.140      0.000       0.001       0.001
sigma2      6.151e-05   2.63e-06     23.405      0.000    5.64e-05    6.67e-05
                             Regime 1 parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.0014      0.001     -1.863      0.063      -0.003    7.44e-05
sigma2         0.0003   3.06e-05     11.078      0.000       0.000       0.000
                         Regime transition parameters
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
p[0->0]        0.9853      0.004    276.096      0.000       0.978       0.992
p[1->0]        0.0632      0.015      4.178      0.000       0.034       0.093
==============================================================================

Warnings:
[1] Covariance matrix calculated using numerical (complex-step) differentiation.
```